

5	(a) in an electric field, charges (in a conductor) would move	B1	
	no movement of charge so zero field strength		
	or		
	charge moves until $F = 0$ / $E = 0$	B1	[2]
	or		
	charges in metal do not move	(B1)	
	no (resultant) force on charges so no (electric) field	(B1)	
	(b) at P, $E_A = (3.0 \times 10^{-12})/[4\pi\epsilon_0(5.0 \times 10^{-2})^2]$ ($= 10.79 \text{ NC}^{-1}$)	M1	
	at P, $E_B = (12 \times 10^{-12})/[4\pi\epsilon_0(10 \times 10^{-2})^2]$ ($= 10.79 \text{ NC}^{-1}$)	M1	
	or		
	$(3.0 \times 10^{-12})/[4\pi\epsilon_0(5.0 \times 10^{-2})^2] - (12 \times 10^{-12})/[4\pi\epsilon_0(10 \times 10^{-2})^2] = 0$		
	or		
	$(3.0 \times 10^{-12})/[4\pi\epsilon_0(5.0 \times 10^{-2})^2] = (12 \times 10^{-12})/[4\pi\epsilon_0(10 \times 10^{-2})^2]$	(M2)	
	fields due to charged spheres are (equal and) <u>opposite in direction</u> , so $E = 0$	A1	[3]
	(c) potential $= 8.99 \times 10^9 \{(3.0 \times 10^{-12})/(5.0 \times 10^{-2}) + (12 \times 10^{-12})/(10 \times 10^{-2})\}$	C1	
	$= 1.62 \text{ V}$	A1	[2]
	(d) $\frac{1}{2}mv^2 = qV$		
	$E_K = \frac{1}{2} \times 107 \times 1.66 \times 10^{-27} \times v^2$	C1	
	$qV = 47 \times 1.60 \times 10^{-19} \times 1.62$	C1	
	$v^2 = 1.37 \times 10^8$		
	$v = 1.2 \times 10^4 \text{ ms}^{-1}$	A1	[3]

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